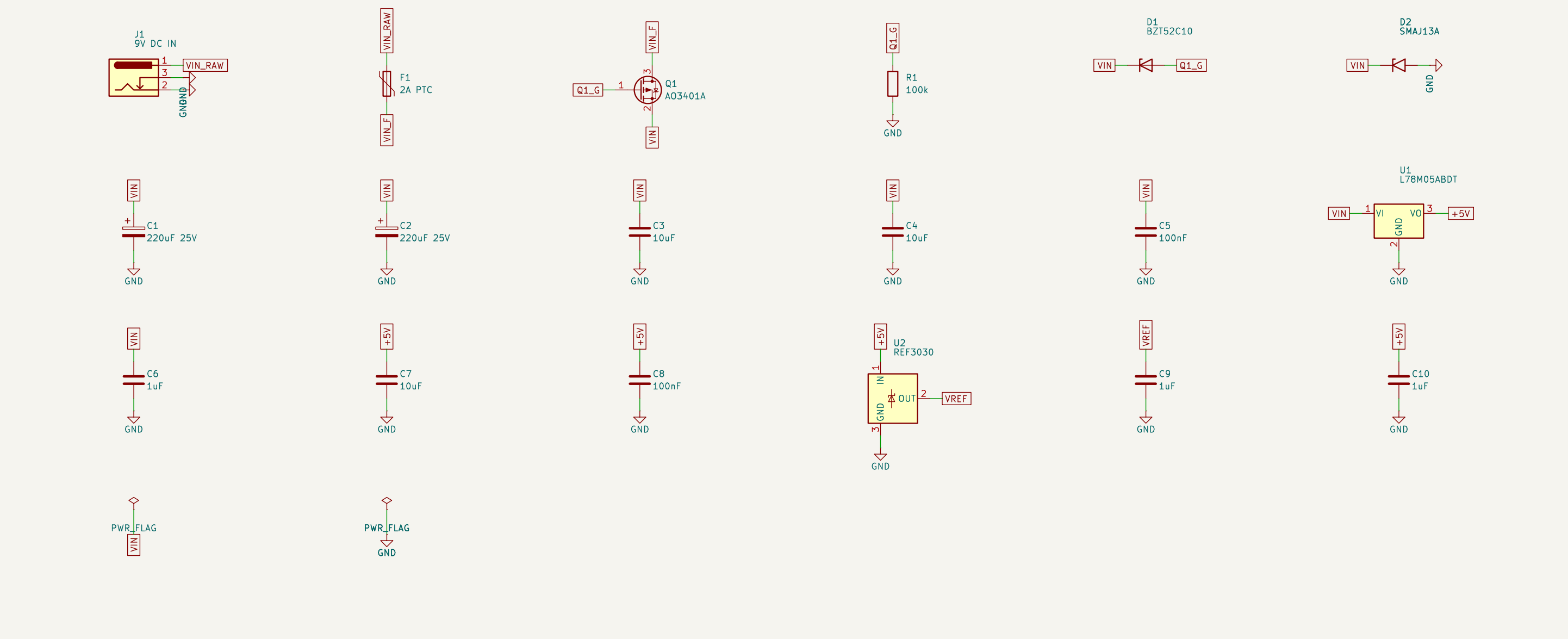
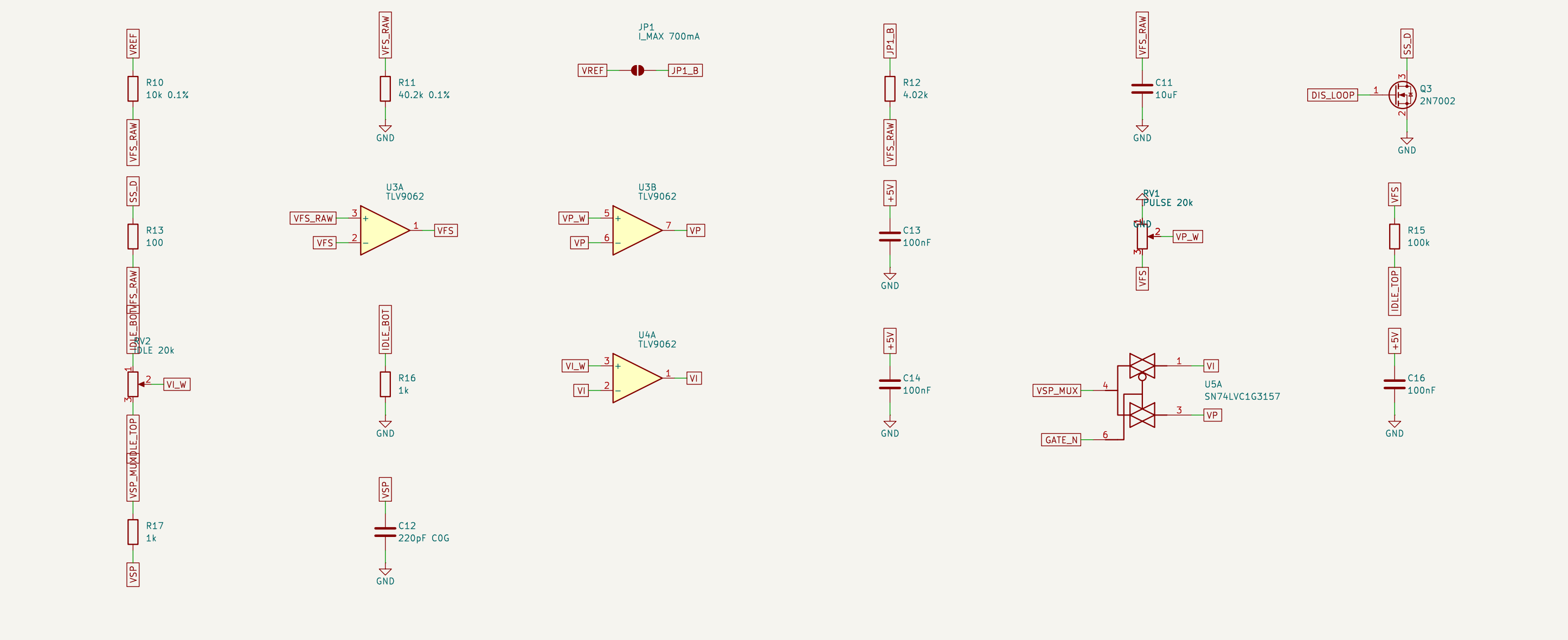
1 POWER INPUT, 5 V AND 3.000 V REFERENCE

9 V / >=1.5 A desktop adapter (5.5/2.1 mm, centre +).
Q1 = reverse-polarity protection, F1 = resettable fuse, D2 = TVS.


J1
9V DC IN
VIN_RAW
GND
F1
2A PTC
VIN_F
Q1
AO3401A
Q1_G
VIN
R1
100k
GND
D1
BZT52C10
D2
SMAJ13A
U1
L78M05ABDT
PWR_FLAG

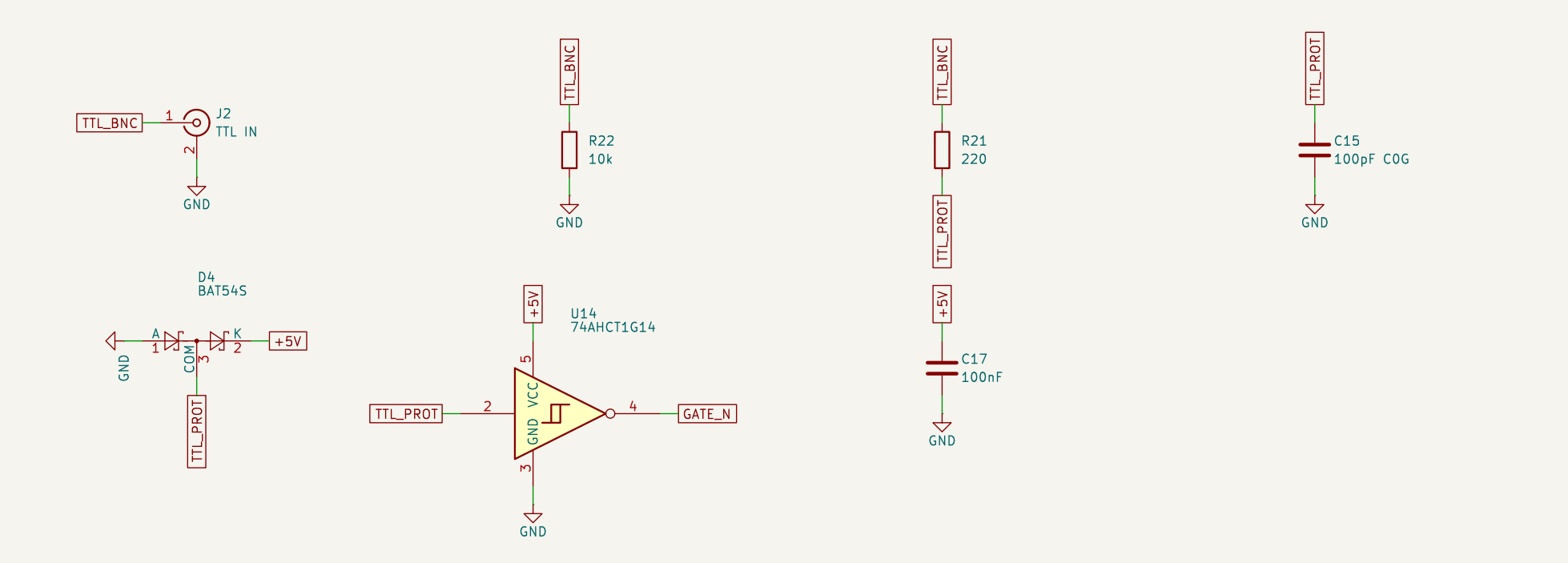
2 SETPOINT, SOFT START AND TTL GATING (scale 4 V/A)

VFS = 2.40 V = 600 mA full scale (JP1 closed: 2.80 V = 700 mA).
C11 + R10||R11 give an 80 ms soft start; Q3 discharges it when disabled.
U5 selects PULSE (TTL high) or IDLE (TTL low, 5..104 mA).


VREF
R10
10k 0.1%
VFS_RAW
R11
40.2k 0.1%
GND
JP1
I_MAX 700mA
JP1_B
R12
4.02k
C11
10uF
SS_D
DIS_LOOP
Q3
2N7002
U3A
U3B
R17
1k
VSP
C12
220pF C0G

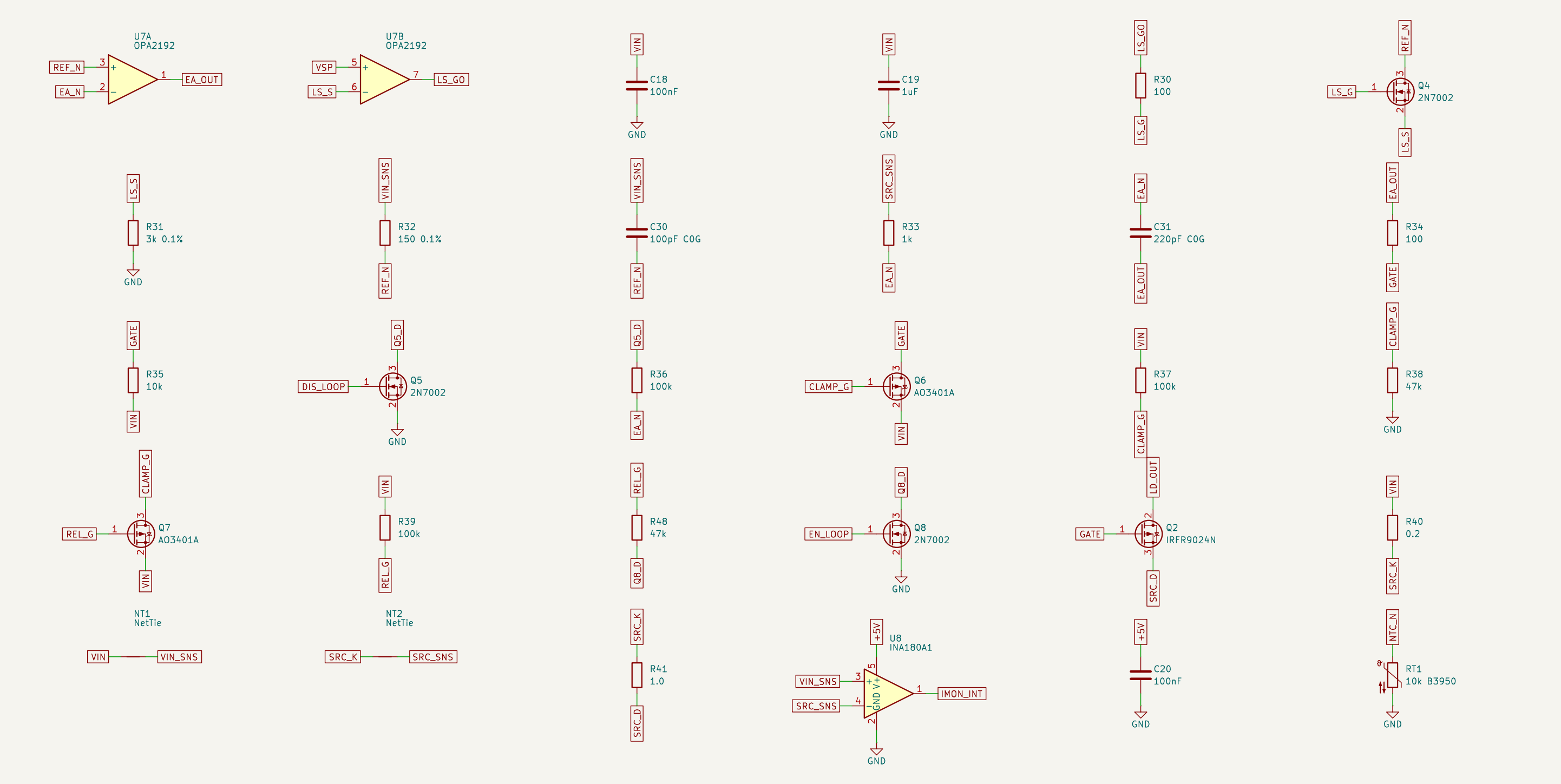
3 TTL INPUT (BNC, 10 k to GND, 3.3/5 V logic)

Schmitt trigger with TTL thresholds; unconnected input = IDLE.


TTL_BNC
J2
TTL IN
GND
R22
10k
R21
220
TTL_PROT
C15
100pF C0G
D4
BAT54S

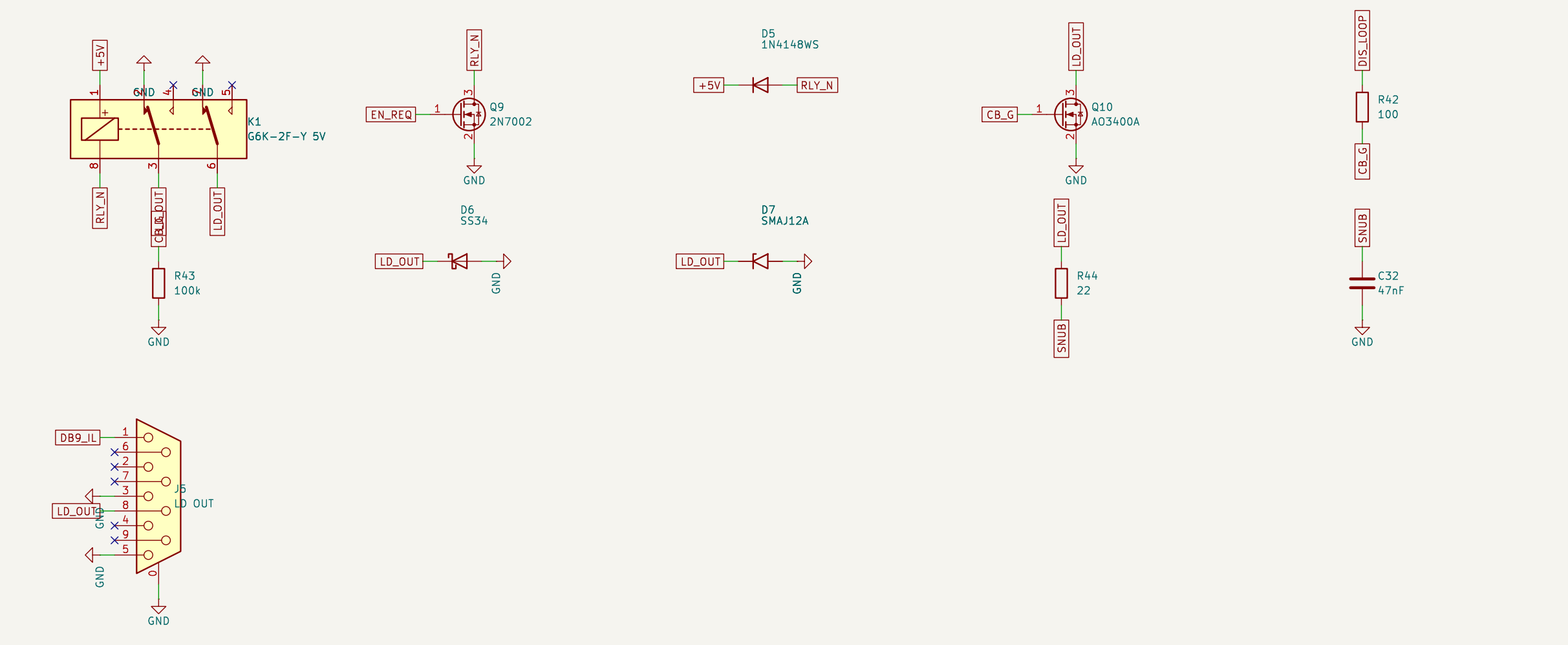
4 HIGH-SIDE CURRENT SOURCE (laser cathode at GND)

I_LD = VSP x (R32/R31) / R40 = VSP / 4 V/A.
U7A error amp senses R40 directly (Kelvin via NT1/NT2); R41 degenerates Q2.
Q6 clamps Q2 off unless EN_LOOP is high (fail-safe, Q7/Q8 release it).
Q5 forces U7A high while disabled so the integrator cannot wind up.


U7A
OPA2192
REF_N
EA_N
EA_OUT
U7B
OPA2192
VSP
LS_S
LS_GO
C18
100nF
C19
1uF
VIN
GND
R30
100
LS_G
Q4
2N7002
REF_N
VIN_SNS
SRC_K
SRC_SNS
R41
1.0
SRC_D
U8
INA180A1
IMON_INT
C20
100nF
RT1
10k B3950

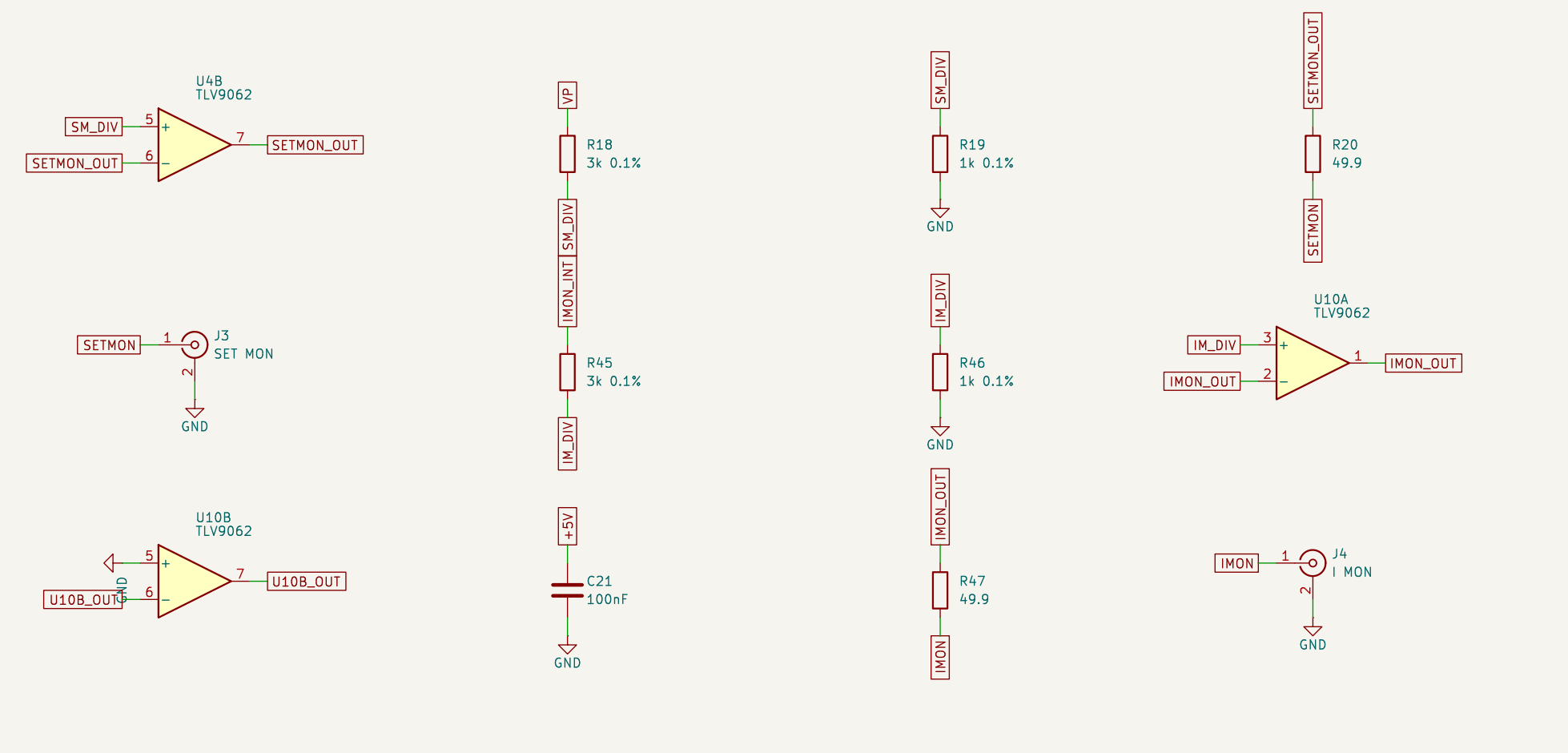
5 OUTPUT, SHORTING RELAY AND CROWBAR

K1 NC contacts short the laser whenever the driver is not armed or unpowered.
Q10 crowbar shorts it within 1 us of a fault; D6/D7 reverse and surge clamp.
DB9 follows Thorlabs LD-driver 'CG' pinout: 8 = anode, 3 = cathode/GND, 1-5 interlock.


+5V
GND
K1
G6K-2F-Y 5V
RLY_N
LD_OUT
EN_REQ
Q9
2N7002
D5
1N4148WS
Q10
AO3400A
CB_G
DIS_LOOP
R42
100
D6
SS34
D7
SMAJ12A

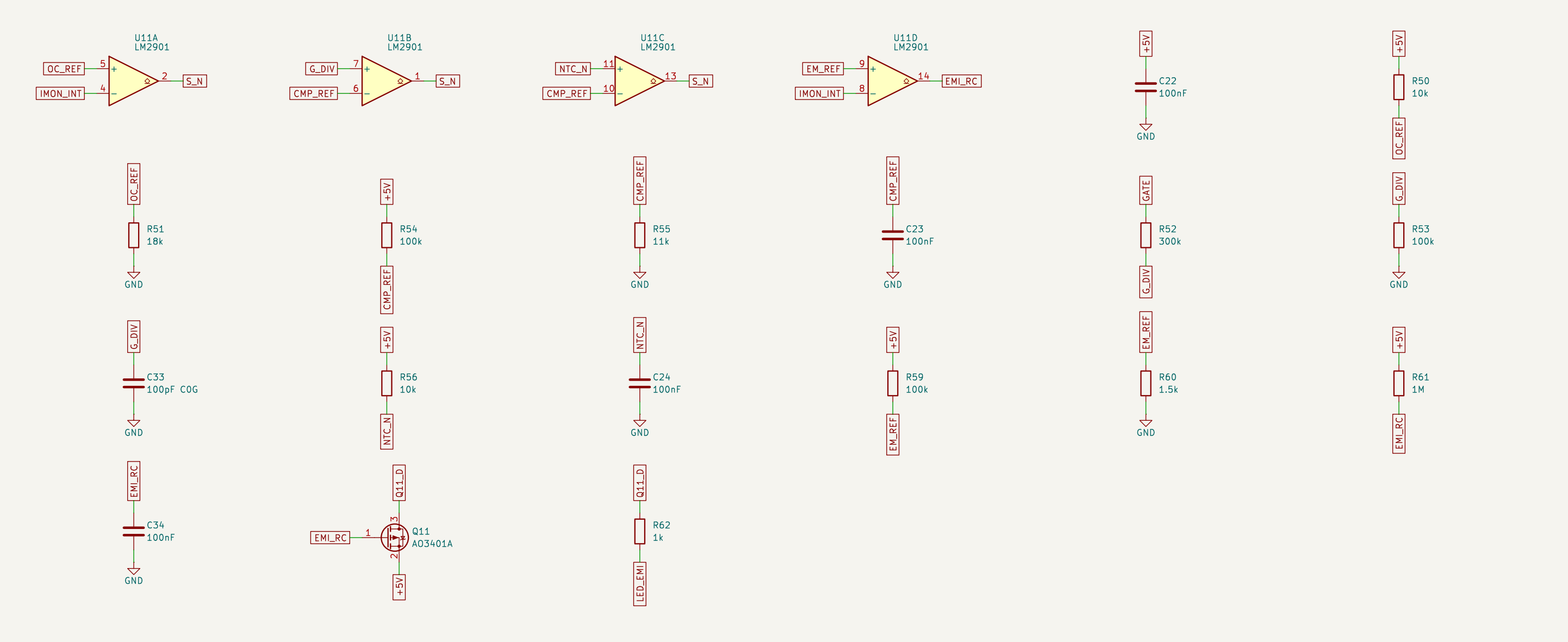
6 CURRENT AND SETPOINT MONITORS (1 V/A into high-Z)

I MON from INA180 (4 V/A) / 4; SET MON shows the PULSE setpoint / 4.


U4B
TLV9062
SM_DIV
SETMON_OUT
VP
R18
3k 0.1%
R19
1k 0.1%
GND
R20
49.9
SETMON
U10A

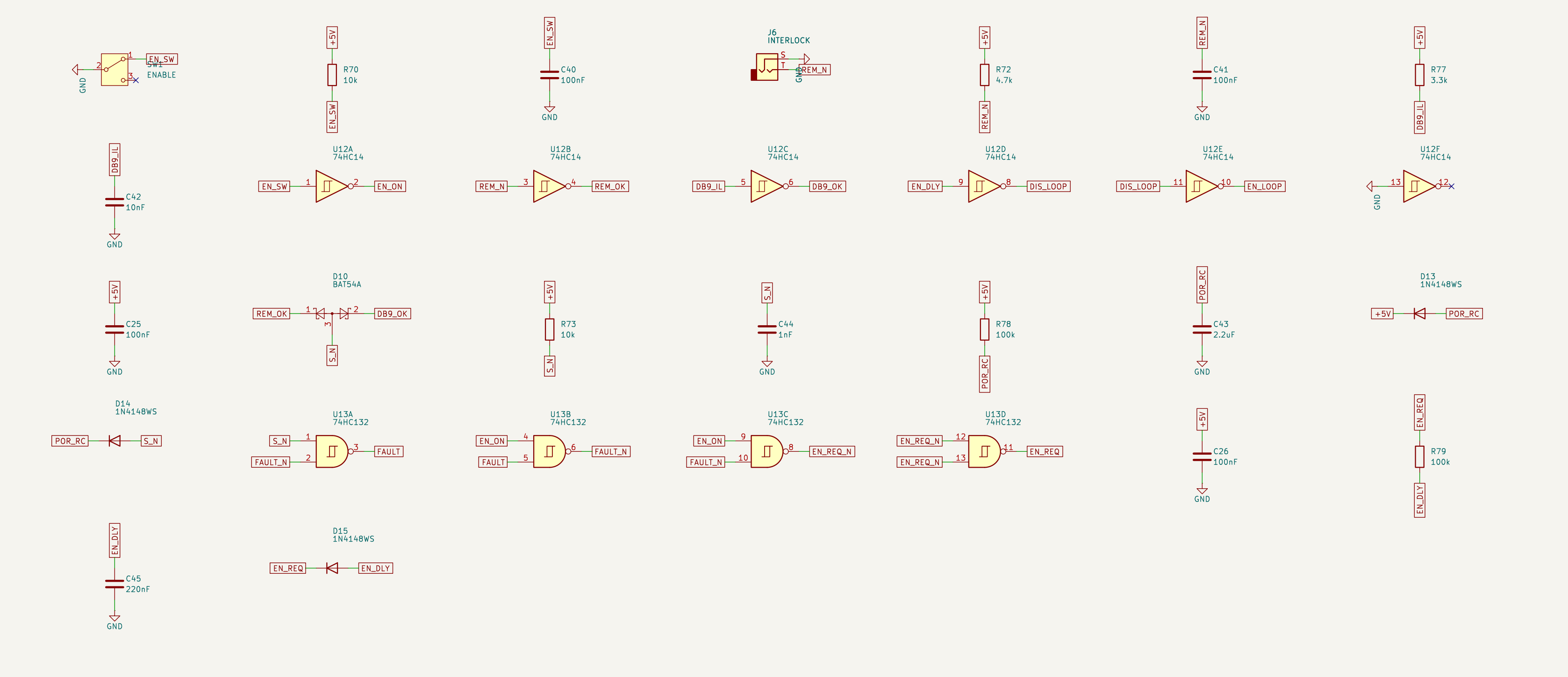
7 PROTECTION COMPARATORS

A: over-current > 800 mA. B: compliance / open laser (gate < 2 V for > 10 us).
C: Q2 heatsink > 85 C (NTC RT1). D: emission > 19 mA lights the EMISSION LED for ~0.15 s.

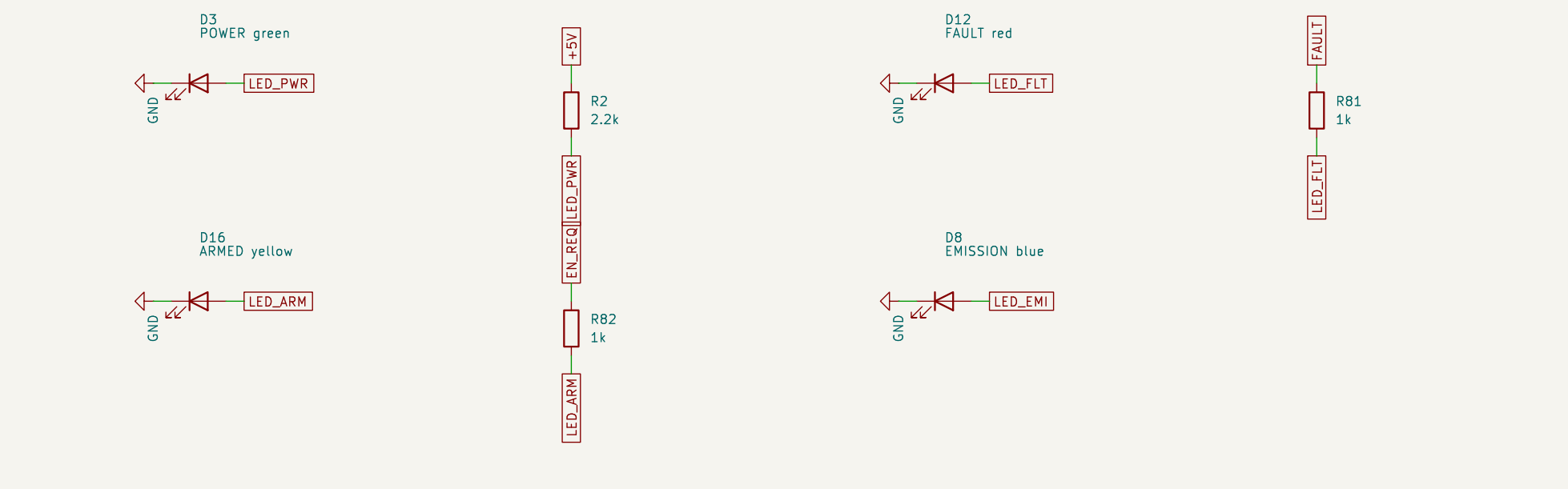

U11A
LM2901
OC_REF
IMON_INT
S_N
U11B
LM2901
G_DIV
CMP_REF
U11C
LM2901
NTC_N
U11D
LM2901
EM_REF
EMI_RC
+5V
C22
100nF
GND
R50
10k

8 INTERLOCKS, FAULT LATCH AND ENABLE SEQUENCING

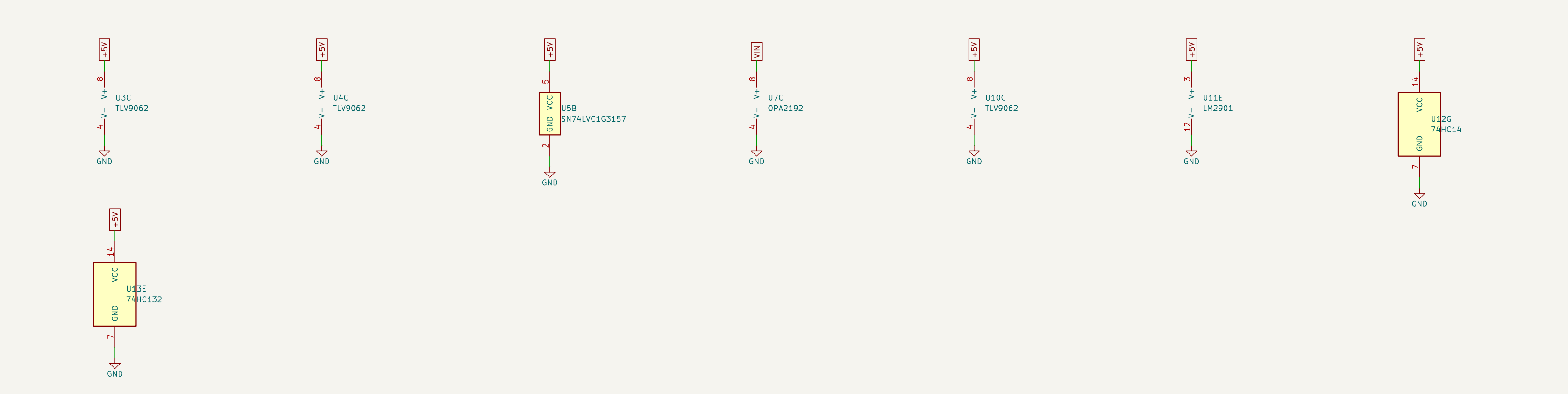
Any fault, open interlock or power-up sets the FAULT latch; reset by switching ENABLE off.
EN_REQ arms relay K1 at once; EN_LOOP releases the current source ~17 ms later.


SW1
ENABLE
EN_SW
GND
+5V
R70
10k
C40
100nF
J6
INTERLOCK
REM_N
R72
4.7k
C41
100nF
R77
3.3k
DB9_IL
U12A
74HC14
U12B
74HC14
U12C
74HC14
U12D
74HC14
U12E
74HC14
U12F
74HC14

9 FRONT-PANEL INDICATORS



10 IC POWER PINS



11 MOUNTING HOLES

CLASS 4 LASER – read README safety section before use
Laser: DTR BDR–209 405 nm (500 mA nominal, 600 mA max), CATHODE AT GND
Generated from tools/design.py by tools/gen_sch.py
Ion–trap lab photoionisation 405 nm driver

Sheet: /
File: ld405–driver.kicad_sch

Title: 405 nm laser diode pulse driver (BDR–209)

Size: A0	Date: 2026–10–28	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1